

atavide-lite: simplified atavistic metagenome processing

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Summary

atavide lite is a modular metagenomics processing pipeline designed to handle complex, multi-sample, multi-technology datasets. Instead of a single monolithic workflow, it separates processing into discrete, independent steps, allowing users to run only what they need, retry failed steps without restarting the entire workflow, and optimise resource use for their specific environment. The pipeline integrates both read-based and assembly-based approaches, and supports both paired-end short reads and long-read sequencing data.

Complete documentation for [atavide lite](#) is available on [read the docs](#).

Statement of need

All environments are impacted by microbes and viruses. Over the last two decades great strides have been made in exploring what microbes and viruses are present in a wide swath of environments, and understanding what they are doing, and how they contribute to biogeochemical cycles, health, or disease.

Driven by decreases in the costs of DNA sequencing, new DNA sequencing technologies, and the availability of abundant and cheap computing, metagenomics has emerged as one of the most complete methods to understand the microbes and viruses in any environment. Typically, metagenomics processing starts by size partitioning the microbial components to focus on the bacteria or viruses, and then extracting all the DNA present in the sample. Advanced computational techniques are then applied to explore which microbes and viruses are present (their taxonomy) and the functions that they are performing (by annotating their genes). By leveraging the additional information that multiple samples provide, metagenomic binning strives to reconstruct complete, or near-complete, microbial and viral genomes from complex samples.

atavide_lite performs all the computational processing of metagenomics sequence data, starting with raw reads from a DNA sequencing machine, we end up with annotated reads and reassembled genomes.

State of the field

Numerous metagenomics pipelines exist, from web-based systems like MG-RAST ([Meyer et al., 2008](#)), to a proliferation of command-line pipelines ([Blanco-Míguez et al., 2023](#); e.g. [Clarke](#)

et al., 2019; Garfias-Gallegos et al., 2022; Laudadio et al., 2019; Lu et al., 2022; Roach et al., 2024; Tyagi & Katara, 2024; Walker et al., 2022). Workflow management systems like Nextflow (Di Tommaso et al., 2017) and Snakemake (Köster & Rahmann, 2012) streamlined the creation of pipelines for bioinformatics analysis, including metagenomics.

Most metagenomics pipelines process sequence data in the same way. Each starts with quality control of the raw fastq files retrieved from the DNA sequencing facility. Quality control includes removing low quality reads, removing adapters and linkers that remain in the sequences, and removing overly short or long reads. Next, if the sample is a host-associated sample, the pipelines remove reads that map to the host genome. At this point, the pipelines often branch into read-based analysis and contig-based analysis. For read-based analysis, individual reads are annotated for taxonomy and function. The number of reads that map to each category is a proxy for the abundance of that category in the original sample, and so mapped reads are counted and their abundance normalised. For contig-based analysis, the contigs are grouped or binned into sets of contigs that likely originated from the same organism, and then those contigs are annotated.

The continued decrease in the cost of DNA sequencing, together with the ever increasing sequencing throughput and computing power, mean that these days we routinely generate metagenomics datasets with hundreds or thousands of samples, and each sample needs to be processed through this pipeline.

Managing this many samples across real-world, large-scale deployments reveal two persistent problems:

1. Fragility at scale: When processing thousands of datasets, there are multiple failure points in each metagenomics analysis.
2. Poor portability across HPC environments: Optimising for one cluster often breaks performance or compatibility on another, especially when each platform has idiosyncratic storage, job scheduling, and runtime constraints.

For example, when processing thousands of samples from the Sequence Read Archive (SRA) (Leinonen et al., 2011), we found that some samples failed to download, and we had to interrupt the pipeline to complete the downloads, fix the issue, and restart the pipeline. When processing our own data, corrupt or inconsistent sequence files (e.g. fastq files with different length sequences or quality scores), can crash a run. Similarly, when using large computations, such as comparing sequence reads to a database of reference sequences using MMSeqs (Steinegger & Söding, 2017), the computation occasionally times out because of limitations of the compute environment, the compute fails because of unforeseen software incompatibility, or the computation fails for a variety of other reasons. Although pipelines streamline and seemingly simplify the process, our experience showed that complex pipelines fail with different samples at different states, obfuscating the precise cause of the failure.

Moreover, each of the computational platforms we use has nuances for the most efficient computational processing. For example, our institutional HPC has a very fast local disk which is available from a non-standard location and is only available to compute nodes. The Australian National Computational Infrastructure's Gadi system does not support large array jobs. The Pawsey Supercomputing Centre's Setonix system regularly deletes files on their "/scratch" disk, but has a large S3 storage system that is available to all compute nodes but from a specific name.

atavide_lite addresses these challenges by replacing an "all-or-nothing" execution model of most computational pipelines with independent, platform-optimised steps. We provide generic scripts that will run on most clusters, plus tuned configurations for the HPC systems we use, and encourage users to adapt them to their own environments. We provide an issue form that allows users to request configurations for specific systems, and provide documentation to enable agentic-AI development of the pipelines tailored to any system. This design improves fault-tolerance, eases debugging, and enables efficient use of heterogeneous compute infrastructures.

Software design

The overall atavide_pipeline is presented in [Figure 1](#).

Design decisions

When designing atavide_lite we have attempted to make code that is re-usable on any system, and to simplify the processing steps. Specific design decisions we took include:

User defined variables when possible.

We use a generic DEFINITIONS.sh file to define some common variables, and that should reside in each analysis directory and becomes part of the analysis.

Similar analysis for single-end and paired-end reads.

Although different instruments and technologies vary drastically, once the sequences themselves are generated, the analysis is usually quotidian. Wherever possible, we use the same analysis for all sequence read data.

Using array jobs when possible.

Array scheduling on HPC systems allows users and administrators to issue a single command to control many tens or hundreds of jobs. Each cluster has different limits on how many jobs may be a part of an array, and this will be determined by your systems administrators, but generally more than 100 jobs are permissible in a single array.

Leverage HPC scheduler controls when possible.

Most HPC systems have a controller that can start queued jobs based on dependencies. When possible, we queue all of the jobs at once, but allow the controller to start the dependencies if, and only if, the preceding job successfully completed. Depending on the specific HPC setup, dependent jobs may be marked as unable to complete or removed from the queue. If a job fails, the output and the logs will be clear where the failure point occurred so that the specific failure issue can be fixed before the computation is restarted.

Computational steps in the pipeline.

Before the data is processed, there are two steps that the user needs to perform. First, we create a DEFINITIONS.sh file in the working directory. This file defines a few variables that are used throughout the processing. First, we define the file ending for the fastq and fasta files. These endings are typically sequence instrument vendor specific (e.g. MGI-generated fastq files may have different file endings from Illumina-generated fastq files), and are trimmed from later files. We define a sample name which covers the whole dataset, and we define the location of the host genome sequence and where to save the host-related and not host-related sequences. Note the last three can be omitted if host mapping is not required.

Next, we create an R1_reads.txt or reads.txt file and either a NUM_R1_READS or NUM_READS variable. The R1_reads.txt file is required for processing paired end reads, while the single end reads uses reads.txt. The variables are really just a simple convenience and are not strictly necessary for processing the jobs, but make it easier to submit array jobs.

We start with quality control using *fastp* ([Chen et al., 2018](#)) to remove any sequence with too many N bases, remove adapters, and reads that are too short. For ONT data, our defaults are two N's and a minimum length of 50 bp, while for short-read data, our defaults are 1 N base and a minimum length of 100 bp.

The second step, if required, is to remove host DNA. *atavide_lite* maps the reads that pass quality control to the host genome using *minimap2* (H. Li, 2018), and then uses *samtools* (H. Li et al., 2009) to filter the reads. Sequences that do not match the flag 3588 – and thus represent mapped reads which pass the platform/vendor quality check, are not PCR or optical duplicates, and are primary alignments – are filtered as mapping to the host genome. For short reads, additional *samtools* flags are used to identify R1 reads (matching flag 65) and R2 reads (matching flag 129). Reads that do not match the flag 3584 – and thus represent unmapped reads which pass the platform/vendor quality check, are not PCR or optical duplicates, and are primary alignments – are filtered as unmapped reads (H. Li et al., 2009) and again, short reads were separated using flags that match 77 for R1 reads and 141 for R2 reads. The matching and unmapping reads are stored separately, although currently we do not use the reads that match the host genome for any downstream analysis.

Note that the host genome can be any genome or multi-fasta record that the user wants excluded from subsequent analysis, and is defined in the aforementioned `DEFINITIONS.sh` file.

Reads that do not map to the host genome are converted to fasta format, and then mapped against the UniRef100 database (Suzek et al., 2007) using the easy-taxonomy workflow built into *MMseqs2*.

MMseqs2 easy-taxonomy parameters were optimised by the authors for a balance of sensitivity and speed and include a 7-mer double-match prefilter with compositional bias correction, low-complexity masking with TANTAN, and default similarity thresholds ($-k$ -score ~ 95) that generate sufficient similar k -mers for accurate detection while maintaining computational efficiency. Candidate hits are further filtered by fast ungapped alignment and refined with vectorised Smith-Waterman alignments, after which taxonomic labels are assigned using a lowest common ancestor (LCA) approach described in more detail in their paper and on their website (Steinegger & Söding, 2017). The taxonomies are reformatted with *taxonkit* (Shen & Ren, 2021) and merged into a single table.

The functional annotations were enriched by mapping the UniProt IDs from the *MMseq2* output directly to proteins associated with BV-BRC subsystems (Olson et al., 2023) (this was a 1:1 mapping and did not require thresholds/cutoffs). The number of reads that *MMseqs2* reported mapped to each protein was counted, and the total was divided by the number of mapped reads to normalise the read counts.

For the assembly-based approaches, the metagenomes were assembled using *megahit* (D. Li et al., 2015) which by default employs a succinct *de Bruijn* graph approach with multiple k -mer sizes (21 to 141 in steps of 20), automatic memory optimisation, and conservative heuristics for contig pruning to balance assembly sensitivity, contiguity, and computational efficiency. Metagenome-assembled genomes (MAGs) (Mallawaarachchi et al., 2024) were reconstructed using *VAMB* (Nissen et al., 2021), a variational autoencoder-based approach that jointly encodes k -mer composition and differential coverage profiles of contigs into a low-dimensional latent space. By learning these compressed representations, *VAMB* effectively separates contigs originating from different microbial genomes and clusters them using a medoid-based algorithm. This deep learning approach has been shown to improve bin purity and completeness compared to conventional binning methods such as *MetaBAT* (Kang et al., 2019) or *CONCOCT* (Alneberg et al., 2014), particularly in complex metagenomic datasets (Papudeshi et al., 2017).

Database selection

We routinely use the larger UniRef100 database for our computations because it contains more protein sequences, and is thus likely to find more matches. However, the trade-off is that it takes significantly longer to compare sequences to the UniRef100 database than the smaller UniRef50 database (protein sequences that are clustered at 50% identity). At the moment UniRef100 is 275G while UniRef50 is 29G. For example, in one recent comparison using the same metagenomic dataset, the UniRef50 search took 503 seconds using 64 threads (8 minutes

182 23 seconds). The equivalent UniRef100 search took 9,846 seconds (2 hours 44 minutes 6
183 seconds) using the same 64 threads.

184 At both genus and family taxonomic levels, the log-transformed UniRef50 and UniRef100-derived
185 profiles were broadly concordant. The sample-level totals were almost perfectly correlated
186 between the two database choices: the two approaches largely agreed on which samples had
187 higher or lower overall assignment levels. However, the two approaches were not identical.
188 UniRef100 assigned more total counts and, in some cases, appeared to resolve reads that were
189 left at broader or more ambiguous taxonomic levels in the UniRef50-derived result.

190 **Research impact statement**

191 We routinely use `atavide_lite` to process metagenomics datasets from a wide range of
192 environments and hosts. For example, we studied how microbes migrate through complex
193 environments (Grigson et al., 2026) which did not require any host genome removal steps. In
194 contrast, our studies on the microbial and viral communities of the CF airways [Carlson-Jones
195 et al. (2026); Goddard2026-ee] and the causes of an undiagnosed illness in Central Africa
196 (Wawina-Bokalanga et al., 2026) both required the removal of human genome contamination.
197 Meanwhile, when we studied the microbes on the surface of elasmobranchs (Doane et al., 2026)
198 we eliminated known shark genomes from our analysis. In each of these very different sampling
199 schemes, sequenced with different technologies, we used the same underlying pipeline.

200 **AI usage disclosure**

201 We widely use agentic AI in code development. Our typical development pipeline now includes
202 initial project scoping with ChatGPT leading to detailed instructions to write code that are
203 provided to either Codex or Claude Code. All commits and pull requests are reviewed with
204 Codex and Copilot in GitHub, and agentic AI is used to resolve issues. We provide detailed
205 documentation in our GitHub repository for other users to extend `atavide_lite` to new clusters,
206 and encourage them to use agentic AI in doing so.

207 However, this manuscript was written without AI support.

- 224 Blanco-Míguez, A., Beghini, F., Cumbo, F., McIver, L. J., Thompson, K. N., Zolfo, M.,
225 Manghi, P., Dubois, L., Huang, K. D., Thomas, A. M., Nickols, W. A., Piccinno, G.,
226 Piperni, E., Punčochář, M., Valles-Colomer, M., Tett, A., Giordano, F., Davies, R., Wolf,
227 J., ... Segata, N. (2023). Extending and improving metagenomic taxonomic profiling
228 with uncharacterized species using MetaPhlAn 4. *Nat. Biotechnol.*, 41, 1633–1644.
229 <https://doi.org/10.1038/s41587-023-01688-w>
- 230 Carlson-Jones, J. A. P., Goddard, T. R., Papudeshi, B., Mallawaarachchi, V., Whiteson, K.
231 L., Warner, M. S., Morton, J. M., Jersmann, H. P. A., & Edwards, R. A. (2026). DNA
232 sequencing for microbial surveillance in cystic fibrosis airways: Advances, challenges, and
233 clinical translation. *Clin. Microbiol. Rev.*, e00352-25. <https://doi.org/10.1128/cmr.00352-25>
234
- 235 Chen, S., Zhou, Y., Chen, Y., & Gu, J. (2018). Fastp: An ultra-fast all-in-one FASTQ
236 preprocessor. *Bioinformatics*, 34(17), i884–i890. <https://doi.org/10.1093/bioinformatics/bty560>
237
- 238 Clarke, E. L., Taylor, L. J., Zhao, C., Connell, A., Lee, J.-J., Fett, B., Bushman, F. D.,
239 & Bittinger, K. (2019). Sunbeam: An extensible pipeline for analyzing metagenomic
240 sequencing experiments. *Microbiome*, 7(1), 46. <https://doi.org/10.1186/s40168-019-0658-x>
241
- 242 Di Tommaso, P., Chatzou, M., Floden, E. W., Barja, P. P., Palumbo, E., & Notredame, C.
243 (2017). Nextflow enables reproducible computational workflows. *Nat. Biotechnol.*, 35, 316.
244 <https://doi.org/10.1038/nbt.3820>
- 245 Doane, M., Martin, B., Kerr, E. N., Dinsdale, E. A., Guida, L., & Kyne, P. (2026). Functional
246 redundancy and anaerobic metabolism characterize the skin microbiome of a critically
247 endangered sawfish. *bioRxiv*, 2026.03.05.709720. <https://doi.org/10.64898/2026.03.05.709720>
248
- 249 Flinders University. (2021). *Deep thought (HPC)*. Flinders University. <https://doi.org/10.25957/FLINDERS.HPC.DEEPTHOUGHT>
250
- 251 Garfias-Gallegos, D., Ziri6n-Mart6nez, C., Bustos-D6az, E. D., Arellano-Fern6ndez, T. V.,
252 Lovaco-Flores, J. A., Espinosa-Jaime, A., Avelar-Rivas, J. A., & S6lem-M6jica, N. (2022).
253 Metagenomics bioinformatic pipeline. *Methods Mol. Biol.*, 2512, 153–179. https://doi.org/10.1007/978-1-0716-2429-6_10
254
- 255 Grigson, S. R., Hutton, A. L. K., Carlson-Jones, J. A. P., Paterson, J. S., Giles, S. K., Harker, C.
256 M., Grigson, A. K., Annells, A. N., Roach, M. J., Speck, P. G., Edwards, R. A., & Mitchell,
257 J. G. (2026). Co-migration of hundreds of microbial species over metres drives selection
258 and promotes non-motile hitchhikers. *Nat. Commun.* <https://doi.org/10.1038/s41467-026-72998-1>
259
- 260 Kang, D. D., Li, F., Kirton, E., Thomas, A., Egan, R., An, H., & Wang, Z. (2019). MetaBAT
261 2: An adaptive binning algorithm for robust and efficient genome reconstruction from
262 metagenome assemblies. *PeerJ*, 7(e7359), e7359. <https://doi.org/10.7717/peerj.7359>
- 263 K6ster, J., & Rahmann, S. (2012). Snakemake—a scalable bioinformatics workflow engine.
264 *Bioinformatics*. <https://academic.oup.com/bioinformatics/article-abstract/28/19/2520/290322>
265
- 266 Laudadio, I., Fulci, V., Stronati, L., & Carissimi, C. (2019). Next-generation metagenomics:
267 Methodological challenges and opportunities. *OMICS*, 23, 327–333. <https://doi.org/10.1089/omi.2019.0073>
268
- 269 Leinonen, R., Sugawara, H., Shumway, M., & International Nucleotide Sequence Database
270 Collaboration. (2011). The sequence read archive. *Nucleic Acids Res.*, 39, D19–21.
271 <https://doi.org/10.1093/nar/gkq1019>

- Li, D., Liu, C.-M., Luo, R., Sadakane, K., & Lam, T.-W. (2015). MEGAHIT: An ultra-fast single-node solution for large and complex metagenomics assembly via succinct de bruijn graph. *Bioinformatics*, 31(10), 1674–1676. <https://doi.org/10.1093/bioinformatics/btv033>
- Li, H. (2018). Minimap2: Pairwise alignment for nucleotide sequences. *Bioinformatics*, 34(18), 3094–3100. <https://doi.org/10.1093/bioinformatics/bty191>
- Li, H., Handsaker, B., Wysoker, A., Fennell, T., Ruan, J., Homer, N., Marth, G., Abecasis, G., Durbin, R., & 1000 Genome Project Data Processing Subgroup. (2009). The sequence alignment/map format and SAMtools. *Bioinformatics*, 25(16), 2078–2079. <https://doi.org/10.1093/bioinformatics/btp352>
- Lu, J., Rincon, N., Wood, D. E., Breitwieser, F. P., Pockrandt, C., Langmead, B., Salzberg, S. L., & Steinegger, M. (2022). Metagenome analysis using the kraken software suite. *Nat. Protoc.*, 17, 2815–2839. <https://doi.org/10.1038/s41596-022-00738-y>
- Mallawaarachchi, V., Wickramarachchi, A., Xue, H., Papudeshi, B., Grigson, S. R., Bouras, G., Pahl, R. E., Kaphle, A., Verich, A., Talamantes-Becerra, B., Dinsdale, E. A., & Edwards, R. A. (2024). Solving genomic puzzles: Computational methods for metagenomic binning. *Briefings in Bioinformatics*, 25(5), bbae372. <https://doi.org/10.1093/bib/bbae372>
- Meyer, F., Paarmann, D., D'Souza, M., Olson, R., Glass, E. M., Kubal, M., Paczian, T., Rodriguez, A., Stevens, R., Wilke, A., Wilkening, J., & Edwards, R. A. (2008). The metagenomics RAST server - a public resource for the automatic phylogenetic and functional analysis of metagenomes. *BMC Bioinformatics*, 9, 386. <https://doi.org/10.1186/1471-2105-9-386>
- Nissen, J. N., Johansen, J., Allesøe, R. L., Søndersby, C. K., Armenteros, J. J. A., Grønbech, C. H., Jensen, L. J., Nielsen, H. B., Petersen, T. N., Winther, O., & Rasmussen, S. (2021). Improved metagenome binning and assembly using deep variational autoencoders. *Nat. Biotechnol.*, 39(5), 555–560. <https://doi.org/10.1038/s41587-020-00777-4>
- Olson, R. D., Assaf, R., Brettin, T., Conrad, N., Cucinell, C., Davis, J. J., Dempsey, D. M., Dickerman, A., Dietrich, E. M., Kenyon, R. W., Kuscuoglu, M., Lefkowitz, E. J., Lu, J., Machi, D., Macken, C., Mao, C., Niewiadomska, A., Nguyen, M., Olsen, G. J., ... Stevens, R. L. (2023). Introducing the bacterial and viral bioinformatics resource center (BV-BRC): A resource combining PATRIC, IRD and ViPR. *Nucleic Acids Res.*, 51(D1), D678–D689. <https://doi.org/10.1093/nar/gkac1003>
- Papudeshi, B., Haggerty, J. M., Doane, M., Morris, M. M., Walsh, K., Beattie, D. T., Pande, D., Zaeri, P., Silva, G. G. Z., Thompson, F., Edwards, R. A., & Dinsdale, E. A. (2017). Optimizing and evaluating the reconstruction of metagenome-assembled microbial genomes. *BMC Genomics*, 18(1), 915. <https://doi.org/10.1186/s12864-017-4294-1>
- Pawsey Supercomputing Research Centre. (2023). *Setonix*. <https://doi.org/10.48569/18SB-8S43>
- Roach, M. J., Beecroft, S. J., Mihindukulasuriya, K. A., Wang, L., Paredes, A., Cárdenas, L. A. C., Henry-Cocks, K., Lima, L. F. O., Dinsdale, E. A., Edwards, R. A., & Handley, S. A. (2024). Hecatomb: An integrated software platform for viral metagenomics. *Gigascience*, 13, giae020. <https://doi.org/10.1093/gigascience/giae020>
- Shen, W., & Ren, H. (2021). TaxonKit: A practical and efficient NCBI taxonomy toolkit. *J. Genet. Genomics*, 48(9), 844–850. <https://doi.org/10.1016/j.jgg.2021.03.006>
- Steinegger, M., & Söding, J. (2017). MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat. Biotechnol.*, 35, 1026–1028. <https://doi.org/10.1038/nbt.3988>
- Suzek, B. E., Huang, H., McGarvey, P., Mazumder, R., & Wu, C. H. (2007). UniRef: Comprehensive and non-redundant UniProt reference clusters. *Bioinformatics*, 23(10),

- 1282–1288. <https://doi.org/10.1093/bioinformatics/btm098>
- Tyagi, S., & Katara, P. (2024). Metatranscriptomics: A tool for clinical metagenomics. *OMICS*, 28, 394–407. <https://doi.org/10.1089/omi.2024.0130>
- Walker, M. E., Simpson, J. B., & Redinbo, M. R. (2022). A structural metagenomics pipeline for examining the gut microbiome. *Curr. Opin. Struct. Biol.*, 75, 102416. <https://doi.org/10.1016/j.sbi.2022.102416>
- Wawina-Bokalanga, T., Makangara-Cigolo, J.-C., Ola-Mpumbe, R., Lokilo, E., Mwakisenda-Tshakotsho, F., Delphine, M., Kahindo, I., Tshonaka-Nkololo, A., Vakaniaki, E. H., Loman, N., Houben, S., Lumembe-Numbi, R., Kinganda-Lusamaki, E., Ponga-Museme, A., Mukota-Nungu, Y., Kumar, A., Meris, M., Wilkinson, S., Colquhoun, R., ... Muyembe-Tamfum, J.-J. (2026). Deciphering the etiology of the 2024 outbreak of undiagnosed febrile illness in panzi, democratic republic of the congo. *Nat. Med.*, 1–9. <https://doi.org/10.1038/s41591-026-04235-7>

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